

ZILONG ZHANG

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EDUCATION

Ph.D. in Machine Learning & Data Science

Aug. 2025 – Now

Department of Mathematics & Statistics / Georgia State University

Atlanta, GA

Master in Statistics

Aug. 2023 – May. 2025

Department of Statistics & Probability / Michigan State University

East Lansing, MI

Bachelor of Financial Mathematics

Sep. 2019 – Jul. 2023

School of Statistics / University of International Business and Economics

Beijing, China

RESEARCH INTERESTS

- **Statistical Machine Learning:** graph neural networks, transfer learning, and statistical theory of deep learning
- **Functional Data Analysis (FDA):** time series, functional principal component analysis, bandwidth selection
- **Bioinformatics / Computational Biology:** single-cell multi-omics integration, RNA–protein prediction, and molecular data analysis

PUBLICATIONS & MANUSCRIPTS

- **Zilong Zhang**, Yi-Ting Hung, Lei Ding, Chi-Kuang Yeh. “Quantifying and Auditing LLM Evaluation via Positive–Unlabeled Learning” *Under Review*, *arXiv:2606.19057*, **2026**.
- Xuan Kan, Zimu Li, Hejie Cui, Yue Yu, Ran Xu, Shaojun Yu, **Zilong Zhang**, Ying Guo, Carl Yang. “R-Mixup: Riemannian Mixup for Biological Networks” *Proceedings of SIGKDD Conference on Knowledge Discovery and Data Mining 2023*, **SIGKDD 2023**.
- Xuan Kan, Wei Dai, Hejie Cui, **Zilong Zhang**, Ying Guo, Carl Yang. “Brain Network Transformer” *Proceedings of the Conference on Neural Information Processing Systems 2022*, **NeurIPS 2022**.

RESEARCH AND OTHER PROJECTS

Adaptive–Refined Clamping for Functional Long-Run Covariance Estimation

Sept. 2025 – Present

- *Project leaders: Dr. Chi-Kuang Yeh (Georgia State University)*
- Proposed ARC, a theoretically grounded bandwidth selection framework for functional time series that combines adaptive estimation, plug-in refinement, and trust-region regularization.

mRNA to Protein Prediction based on Transformer and GNN

Jan. 2024 – Jan. 2026

- *Project Leaders: Dr. Yuying Xie (Michigan State University) and Dr. Lulu Shang (MD Anderson Cancer Center)*
- Developed a novel predictive model based on Transformer and Graph Neural Networks (GNN) to accurately predict protein expression from single-cell RNA data

CRIC Study: Genome-Wide Association of Apparent Treatment-Resistant Hypertension in CKD and ARTH Patients

Sept. 2024 – Dec. 2024

- *Project leader: Dr. Ling Wang (Michigan State University) and Dr. Yuehua Cui (Michigan State University)*
- Analyzed high-dimensional genetic data to identify key genetic variants that contribute to treatment-resistant hypertension in CKD patients, providing potential targets for therapeutic intervention.

Theoretical Research and Application of High Dimensional Variable Screening

Oct. 2022 – May. 2023

- *Project leader: Dr. Wei Xiong (University of International Business and Economics)*
- Combined the application of high-dimensional point map mapping in brain map neural network, to find the most efficient dimensionality reduction method in a specific neural networks situation according to univariate analysis screening method and multivariate analysis screening method.

WORK EXPERIENCE

China Securities Co., Ltd.

Jul. 2021 – Nov. 2021

Intern Investment Manager

Beijing, China

- Assisted in listing work of Beijing Tianzhilang Medical Technology Co., LTD and Golden Sun Grain and Oil Co., LTD.
- Prepared listing application materials, such as industry survey reports and risk assessment reports.
- Planned and executed fundraising activities. Composed market research reports and other review materials.

Chuzhou Longyun Financial Consulting Co. Ltd.

Mar. 2021 – Jun. 2021

General Manager Assistant

Chuzhou, China

- Provided regular financial audit and accounting services to clients.
- Conducted corporate financial audit and due diligence for the corporate.

Bank of Communications Chuzhou Branch

Dec. 2020 – Feb. 2021

Intern Account Manager

Chuzhou, China

- Identified, managed and mitigated accounting risk.
- Set up and maintained credit databases to carry out related creditability assessments and analysis reports.

TECHNICAL SKILLS

Programming: Python, R, C, Sklearn, Numpy, Pandas, PyTorch, Tensorflow, Huggingface

Tools: Git, Linux, Jupyter, VS Code

Skills: Large Language Models, Trustworthy AI, Data Analysis, Statistical Learning Theory

HONORS & AWARDS

- **Second Prize**, Beijing Sub-Region of the 11th National University Students E-commerce “Innovation, Creativity and Entrepreneurship” Challenge Jun. 2021
- **Silver Award**, Challenge Cup Capital University Student Entrepreneurship Competition (Beijing) Sep. 2020

PROFESSIONAL MEMBERSHIPS

- Student Member, **International Chinese Statistical Association(ICSA)** 2026–Present